Zhuoren Li

https://zhuorenli.github.io/ |tongjilzr@tongji.edu.cn | +86 18201975690 | Shanghai, China

Academic and Research Experience

Tongji University

- *Research Assistant* — *Mar 2026 - Present*
- *Ph.D. in Vehicle Engineering (Advisor: Prof. Lu Xiong, Co-advisor: Prof. Jia Hu and Prof. Nan Li)* — *Sep 2019 - Dec 2025*
- *B.E. in Engineering Mechanics* — *Sep 2014 - Jul 2019*

Research Directions

LLM-Enhanced End-to-End Autonomous Driving — *Dec 2024 - Present*

- Scenario understanding for adaptive reward adjustment during E2E RL training
- Physical feasibility enhanced diffusion planning

Control Granularity Improved RL-based Motion Planning — *Dec 2023 – Jun 2025*

- Hierarchical RL for integrated decision-making and motion control
- Hybrid action-based RL using parameterized action space

Safety-Enhanced Reinforcement Learning for Autonomous Driving — *Sept 2022 - Present*

- Prior-knowledge based predictive risk constraint and unsafe behavior correction.
- Multi-Critic mechanism for multi-objective accommodation
- Epistemic uncertainty-based action governor with dynamic control barrier function

Optimization-based Motion Planning — *Sept 2021 – Jun 2023*

- POMDP-based integrated decision-making and motion planning considering prediction uncertainty
- MPC-based Trajectory planning and tracking control

Selected Publications

Journal & Conference Publications:

[1] Dequan Zeng, Zhizhao Ni, **Zhuoren Li***, et al., "Physics-Informed Residual Reinforcement Learning via Expert Prior Knowledge for Safe and Efficient Autonomous Merging," *Neurocomputing*,134631, 2026. [PDF]

[2] Bo Leng, Weiqi Zhang, **Zhuoren Li***, et.al., "Expert Knowledge-driven Reinforcement Learning for Autonomous Racing via Trajectory Guidance and Dynamics Constraints," *Green Energy Intell. Transp.*, 100452, 2026. [PDF]

[3] **Zhuoren Li**, Bo Leng, Lu Xiong, Arno Eichberger, Chao Huang and Jia Hu, "Safety Enhanced Reinforcement Learning for Autonomous Driving: Dare to Make Mistakes to Learn Faster and Better," *IEEE Trans. Intell. Transp. Syst.*, 2026. [PDF]

[4] Guizhe Jin, **Zhuoren Li**, Bo Leng, et al. "Hybrid Action Based Reinforcement Learning for Multi-Objective Compatible Autonomous Driving," *IEEE Trans. Neural Netw. Learn. Sys.*, 2026. [PDF]

[5] Bo Leng, Ran Yu, Chengen Tu, Lu Xiong, Arno Eichberger and **Zhuoren Li***, "Seamless Overtaking Maneuvers for Automated Driving: Integrated Motion Planning Based on Hybrid Model Predictive Control," *IEEE Trans. Ind. Electron*, 2026. [PDF]

[6] Zhiwen Chen, Hanming Deng, **Zhuoren Li***, Huanxi Wen, Guizhe Jin, Ran Yu and Bo Leng*, "HCRMP: A LLM-Hinted Contextual Reinforcement Learning Framework for Autonomous Driving," *Adv. Neural Inf. Process. Syst.(NeruIPS)*, 2025. [PDF]

[7] Ran Yu, **Zhuoren Li***, Lu Xiong, et al., "Uncertainty-Aware Safety-Critical Decision and Control for Autonomous Vehicles at Unsignalized Intersections," *IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2025. [PDF]

[8] Guizhe Jin, **Zhuoren Li**, Bo Leng, Ran Yu, Lu Xiong and Chen Sun, "Multi-Timescale Hierarchical Reinforcement Learning for Unified Behavior and Control of Autonomous Driving," *IEEE Robot. Autom. Lett.*, 10(12): 12772-12779, 2025. [PDF]

[9] Bo Leng, **Zhuoren Li***, Ming Liu, Ce Yang, Yi Luo, Amir Khajepour and Lu Xiong, "Multi-Mode Evasion Assistance Control Method considering Human Driver Operation," *Chin. J. Mech. Eng.*, 2025, 38: 102. [PDF]

[10] Lu Xiong+, **Zhuoren Li**, Danyang Zhong, Puhang Xu and Chen Tang, "Rule-Guidance Reinforcement Learning for Lane Change Decision-making: A Risk Assessment Approach," *Chin. J. Mech. Eng.*, 2025, 38:30. [PDF]

[11] Ruolin Yang, **Zhuoren Li**, Bo Leng, Lu Xiong and Xin Xia, "Convergent Harmonious Decision: Lane Changing in a more Traffic Friendly Way," *IEEE Trans. Intell. Transp. Syst.*, 26(11): 20334-20347, 2025. [PDF]

[12] **Zhuoren Li**, Jia Hu, Bo Leng, Lu Xiong and Zhiqiang Fu, "An Integrated of Decision Making and Motion Planning Framework for Enhanced Oscillation-Free Capability," *IEEE Trans. Intell. Transp. Syst.*, 25(6): 5718-5732, 2024. [PDF]

[13] **Zhuoren Li**, Guizhe Jin, Ran Yu, Bo Leng and Lu Xiong, "Interaction-Aware Deep Reinforcement Learning Approach Based on Hybrid Parameterized Action Space for Autonomous Driving," *SAE Intell. Connected Veh. Symposium*, 2024. [PDF]

[14] **Zhuoren Li**, Lu Xiong, Bo Leng, Puhang Xu and Zhiqiang Fu, "Safe Reinforcement Learning of Lane Change Decision Making with Risk-Fused Constraint," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2023, pp. 1313-1319. [PDF]

[15] **Zhuoren Li**, Lu Xiong Bo Leng. "A Unified Trajectory Planning and Tracking Control Framework for Autonomous Overtaking Based on Hierarchical MPC," in *Proc. IEEE Intell. Transp. Syst. Conf. (ITSC)*, 2022, pp. 937-944. [PDF]

Submitted & In Progress:

[1] **Zhuoren Li**, Yi Zhong, Weiqi Zhang, et al. "Comparison-Based Ordinal Learning for Proactive Driving Risk Assessment." *Accident Analysis & Prevention* (under review) [arXiv]

[2] **Zhuoren Li**, Guizhe Jin, Ran Yu, et al. "A Survey of Reinforcement Learning-Based Motion Planning for Autonomous Driving: Lessons Learned from a Driving Task Perspective." *IEEE Trans. Intell. Transp. Syst.* (under review) [arXiv]

[3] Bo Leng, Ran Yu, **Zhuoren Li*** et al., "Risk-Aware Reinforcement Learning for Autonomous Driving: Improving Safety When Driving through Intersection," *Eng. Appl. Artif. Intel.* (under review) [arXiv]

** corresponding author, + First author of doctoral supervisor*

Selected Projects

Interactive Learning-based Decision Control for Intelligent Vehicles with Roadside Information Enhancement *Apr 2025 - Present*

Chongqing Changan Automobile Co., Ltd., Role: Student Technical Director

- Comprehensive task management.
- Designed an Interaction-aware RL motion planning algorithm combining with game theory guidance.

High-Mobility Motion Planning and Control Research for Chassis-by-wire All-terrain Unmanned Vehicle with Hybrid-steering *Jan 2024 - Present*

National Natural Science Foundation of China, Role: Student Technical Director

- Comprehensive task management.
- Designed an environmental risk characterization algorithm considering negative obstacles.
- Led the Development of a motion planning approach combining hybrid A* and optimization-based methods.

Key Technology of Perception and Control in Cooperative Vehicle-Infrastructure System for Urban Public Transportation *Jan 2023 – Dec 2025*

National Key Research and Development Program of China, Role: Core Participant

- Developed the decision-making and motion planning algorithms for cruise driving and lane change functions in vehicle side, based on the recommended speed from the road side.

Development of Evasion Assistance Algorithm for Emergency Collision Avoidance based on Steer-by-Wire System *Jul 2022 – Jul 2024*

Shanghai Automotive Industry Science and Technology Development Foundation, Role: Student Technical Director

- Comprehensive task management.
- Led the development of a motion control algorithm for emergency collision avoidance.
- Led the simulation validation in Carmaker/Simulink, steer-by-wire system modification on the real vehicle and ground test based on Audesse ECU.

Binary Mixed Traffic Behavior Characteristics and Collaborative Paradigm *Aug 2021 – Jul 2024*

Science and Technology Commission of Shanghai, Role: Core Participant, Role: Work Package Leader

- Motion planning and control of connected automated vehicle according to the road-side guidance.

Research on Vehicle-Road Cooperative Control for Intelligent Public Transportation System *Sep 2021 – Aug 2022*

Shanghai Research Institute for Intelligent Autonomous Systems, - Role: Student Technical Director

- Comprehensive task management.
- Led the development of a signal control algorithm and a vehicle speed optimization algorithm.

Development and Application of Automatic Valet Parking System *Feb 2020 – Mar 2021*

Nanchang Automotive Institute of Intelligence & New Energy, Role: Student Technical Director

- Comprehensive task management.
- Developed an integrated module for hardware and software communication.
- Developed a parking path planning algorithm based on geometric configuration.
- Developed a Stanley-based path tracking algorithm and a feedforward PID-based speed tracking algorithm.

China Future Challenge of Intelligent Vehicles *Mar 2020 – Nov 2020*

Institute of Intelligent Vehicles, Tongji University, Role: Major developer

- Developed a system integration framework inspired by the Baidu Apollo.
- Developed a search-based path planning algorithm for static obstacle avoidance.
- Developed a parking path planning algorithm based on hybrid A*.

Funding/Grant Proposal Writing Experience *Jan 2020 – Mar 2025*

- National Natural Science Foundation of China (NSFC): Excellent Young Scholars Fund, 2025; Key Program, 2024; General Program, 2023; Distinguished Young Scholars Fund, 2022; Young Scholars Grant, 2020.
- Shanghai Municipal People's Government (SMPG), Oriental Excellence Program Youth Project, 2024;
- Shanghai Science and Technology Progress First Award, 2022;
- Ministry of Science and Technology, PRC, National Key Research and Development Program of China, 2021;

Honors and Awards

- SAE International Outstanding Technical Paper Award, SAE ICVS 2024.
- High-Level Academic Poster Award, China SAE Doctoral Student Academic Forum, 2024.
- Outstanding Doctoral Student Scholarship, Tongji University, 2024.
- Vehicle-road-cloud Integrated Autonomous Driving Challenge, Third Prize, 2024.
- Outstanding Individual Award, Institute of Intelligent Vehicles, Tongji University, 2022, 2023, 2024.
- Outstanding Projects Award, Institute of Intelligent Vehicles, Tongji University, 2024.
- World Artificial Intelligence Conference AI Driving Simulation Competition, Third Prize in the University Challenge Competition, 2022

Academic Services

Reviewer

- **Journal Reviewer:** *IEEE Transactions on Intelligent Transportation Systems (TITS), IEEE Transactions on Vehicular Technology (TVT), IEEE Transactions on Intelligent Vehicles (TIV), IEEE Transactions on Transportation Electrification (TTE), IEEE Robotics and Automation Letters (RAL), Transportation Research Part C: Emerging Technologies (TR-C), Engineering Applications of Artificial Intelligence(EAAI), Advanced Engineering Informatics (AEI), Expert Systems With Applications (ESWA), Journal of Intelligent Transportation Systems (JITS), IET Intelligent Transport Systems, Journal of Field Robotics.*
- **Conference Reviewer:** *IEEE IV, ITSC, CVCI, SAE WCX, CCTA*

Mentoring (*serving as the Student Director of the Intelligent Decision Research Group at TJU-IIV since 2021, mentored 5 Ph.D.students, 13 master's students, and several undergraduate students.*)

Ph.D. Students:

- 2025-present: ***Weiqi Zhang*, **E2E RL racing; ***Baoyun Wang***, VLA-based E2E control
- 2023-present: ***Zhiwen Chen***, LLM-enhanced RL for AD.
- 2022-present: ***Peiyuan Fang***, Motion Planning under Off-road Environment; ***Xinrui Zhang***, Cloud-Vehicle Cooperative Planning.

Master Students:

- 2024-present: ***Ran Yu***, Trajectory Prediction and Safe-RL; ***Zhizhao Ni***, LfD-based RL for Merging in Congested Traffic.
- 2023-present: ***Guizhe Jin***, Multi-objective Compatible RL; ***Zhou Sun***, Motion Planning under Off-road Environment.
- 2022-present: ***Yuqin Qi***, Learning-based MPC for Motion Control; ***Yu Che***, Cloud-Vehicle Cooperative Tracking Control.
- 2021-2024: ***Ruolin Yang***, Harmony-enchaned RL; ***Encheng Tu***, Hybrid MPC Motion Planning for Autonomous Overtaking; ***Yizhuo Guan***, Prediction and Control for Emergency Evasion.
- 2020-2023: ***Gesong Shi***, Cooperative Control for Transit Priority.
- 2019-2022: ***Puhang Xu***, Safe-RL Decision-Making; ***Hongyu Xiao***, POMDP Motion Planning; ***Zixuan Qian***, SMPC Motion Planning; ***Jie Gao***: Global Planning.